

Foundations of Overfitting machine learning

[Overfitting machine learning]

1.0 Content Block

Regression In regression analysis, overfitting occurs frequently. As an extreme example, if there are p variables in a linear regression with p data points, the fitted line can go exactly through every point. For logistic regression or Cox proportional hazards models, there are a variety of rules of thumb (e.g. 5–9, 10 and 10–15 — the guideline of 10 observations per independent variable is known as the "one in ten rule"). In the process of regression model selection, the mean squared error of the random regression function can be split into random noise, approximation bias, and variance in the estimate of the regression function. The bias–variance tradeoff is often used to overcome overfit models. With a large set of explanatory variables that actually have no relation to the dependent variable being predicted, some variables will in general be falsely found to be statistically significant and the researcher may thus retain them in the model, thereby overfitting the model. This is known as Freedman's paradox.

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Overfitted models ... are often free of bias in the parameter estimators, but have estimated (and actual) sampling variances that are needlessly large (the precision of the estimators is poor, relative to what could have been accomplished with a more parsimonious model). False treatment effects tend to be identified, and false variables are included with overfitted models. ... A best approximating model is achieved by properly balancing the errors of underfitting and overfitting. Overfitting is more likely to be a serious concern when there is little theory available to guide the analysis, in part because then there tend to be a large number of models to select from. The book *Model Selection and Model Averaging* (2008) puts it this way.

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Increase the complexity of the model: If the model is too simple, it may be necessary to increase its complexity by adding more features, increasing the number of parameters, or using a more flexible model. However, this should be done carefully to avoid overfitting. Use a different algorithm: If the current algorithm is not able to capture the patterns in the data, it may be necessary to try a different one. For example, a neural network may be more effective than a linear regression model for some types of data. Increase the amount of training data: If the model is underfitting due to a lack of data, increasing the amount of training data may help. This will allow the model to better capture the underlying patterns in the data. Regularization: Regularization is a technique used to prevent overfitting by adding a penalty term to the loss function that discourages large parameter values. It can also be used to prevent underfitting by controlling the complexity of the model. Ensemble Methods: Ensemble methods combine multiple models to create a more accurate prediction. This can help reduce underfitting by allowing multiple models to work together to capture the underlying patterns in the data. Feature engineering: Feature engineering involves creating new model features from the existing ones that may be more relevant to the problem at hand. This can help improve the accuracy of the model and prevent underfitting.

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... an underfitted model would ignore some important replicable (i.e., conceptually replicable in most other samples) structure in the data and thus fail to identify effects that were actually supported by the data. In this case, bias in the parameter estimators is often substantial, and the sampling variance is underestimated, both factors resulting in poor confidence interval coverage. Underfitted models tend to miss important treatment effects in experimental settings.

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A function that is overfitted is likely to request more information about each item in the validation dataset than does the optimal function; gathering this additional unneeded data can be expensive or error-prone, especially if each individual piece of information must be gathered by human observation and manual data entry. A more complex, overfitted function is likely to be less portable than a simple one. At one extreme, a one-variable linear regression is so portable that, if necessary, it could even be done by hand. At the other extreme are models that can be reproduced only by exactly duplicating the original modeler's entire setup, making reuse or scientific reproduction difficult. It may be possible to reconstruct details of individual training instances from an overfitted machine learning model's training set. This may be undesirable if, for example, the training data includes sensitive personally identifiable information (PII). This phenomenon also presents problems in the area of artificial intelligence and copyright, with the developers of some generative deep learning models such as Stable Diffusion and GitHub Copilot being sued for copyright infringement because these models have been found to be capable of reproducing certain copyrighted items from their training data.

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Given a data set, you can fit thousands of models at the push of a button, but how do you choose the best? With so many candidate models, overfitting is a real danger. Is the monkey who typed Hamlet actually a good writer?

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Benign overfitting Benign overfitting describes the phenomenon of a statistical model that seems to generalize well to unseen data, even when it has been fit perfectly on noisy training data (i.e., obtains perfect predictive accuracy on the training set). The phenomenon is of particular interest in deep neural networks, but is studied from a theoretical perspective in the context of much simpler models, such as linear regression. In particular, it has been shown that overparameterization is essential for benign overfitting in this setting. In other words, the number of directions in parameter space that are unimportant for prediction must significantly exceed the sample size.

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Remedy The optimal function usually needs verification on bigger or completely new datasets. There are, however, methods like minimum spanning tree or life-time of correlation that applies the dependence between correlation coefficients and time-series (window width). Whenever the window width is big enough, the correlation coefficients are stable and don't depend on the window width size anymore. Therefore, a correlation matrix can be created by calculating a coefficient of correlation between investigated variables. This matrix can be represented topologically as a complex network where direct and indirect influences between variables are visualized. Dropout regularisation (random removal of training set data) can also improve robustness and therefore reduce over-fitting by probabilistically removing inputs to a layer. Pruning is another technique that mitigates overfitting and enhances generalization by identifying a sparse, optimal neural network structure, while simultaneously reducing the computational cost of both training and inference.